

ULETY, Russian Federation

WMO: 308460

Lat: 51.350N

Lon: 112.467E

Elev: 743

StdP: 92.71

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.2	-36.6	0.1	-32.5	-34.0	0.2	-29.8	9.1	-13.4	8.3	-13.9	0.0	260	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.3	17.1	28.2	16.6	26.2	16.0	19.3	26.0	18.4	24.6	17.5	23.3	2.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.5	21.5	16.3	12.7	20.7	15.4	11.9	20.0	58.5	26.1	55.4	24.6	52.5	23.3	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.6	-35.3	33.5	3.4	2.4	-37.7	35.2	-39.7	36.6	-41.6	37.9	-44.1	39.6		
(5)			-35.4	21.0	3.4	1.2	-37.8	21.9	-39.8	22.6	-41.7	23.3	-44.2	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-19.4	-14.9	-6.8	2.3	9.7	16.9	19.3	16.6	9.8	0.8	-10.1	-17.1	(6)
(7)		DBStd	14.31	5.80	5.82	6.08	4.59	4.27	3.69	2.72	3.18	3.99	4.78	6.50	6.06	(7)
(8)		HDD10.0	4202	910	698	519	233	58	2	0	1	52	287	602	840	(8)
(9)		HDD18.3	6533	1168	931	778	479	268	68	19	71	257	543	852	1098	(9)
(10)		CDD10.0	805	0	0	0	3	49	209	290	206	46	2	0	0	(10)
(11)		CDD18.3	94	0	0	0	0	1	24	51	18	1	0	0	0	(11)
(12)		CDH23.3	1257	0	0	0	2	68	407	546	209	25	0	0	0	(12)
(13)		CDH26.7	366	0	0	0	0	17	122	174	49	3	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.7	2.8	2.8	3.3	3.2	2.4	2.0	2.0	2.5	2.8	3.0	2.8	(14)
(15)	Precipitation	PrecAvg	351	3	1	3	12	30	62	88	89	36	12	6	5	(15)
(16)		PrecMax	505	9	6	10	41	101	153	166	203	90	37	20	12	(16)
(17)		PrecMin	170	0	0	0	0	3	3	18	24	1	1	0	0	(17)
(18)		PrecStd	81	2	2	3	10	26	34	38	40	23	9	6	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.8	2.0	12.2	21.2	29.1	32.9	33.4	30.5	26.6	17.7	7.4	-1.1	(19)
(20)			MCWB	-5.5	-2.6	3.8	8.6	13.6	17.1	18.7	17.6	14.6	8.4	1.7	-4.5	(20)
(21)		2%	DB	-6.0	-1.1	8.6	17.1	24.8	30.0	30.7	27.9	22.8	14.1	4.3	-4.1	(21)
(22)			MCWB	-8.1	-4.6	1.7	6.7	11.7	16.1	18.1	17.5	12.9	6.2	-0.3	-6.5	(22)
(23)		5%	DB	-8.9	-3.4	5.6	14.0	22.0	27.5	28.5	25.6	19.9	10.9	1.9	-6.2	(23)
(24)			MCWB	-10.6	-6.3	-0.2	5.1	10.4	15.3	17.6	16.7	11.7	4.3	-1.7	-8.1	(24)
(25)		10%	DB	-11.0	-5.7	2.8	11.3	19.3	25.5	26.3	23.4	17.3	8.3	-0.6	-8.1	(25)
(26)			MCWB	-12.3	-8.0	-1.9	3.6	9.1	14.8	17.0	16.1	10.6	3.0	-3.3	-9.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.2	-2.4	4.2	9.0	14.2	19.4	21.0	20.2	15.8	8.7	1.9	-4.1	(27)
(28)			MCDWB	-2.8	1.8	11.9	20.4	26.5	27.7	28.1	26.6	22.9	17.3	7.0	-1.3	(28)
(29)		2%	WB	-8.0	-4.5	1.8	7.0	12.3	17.7	19.7	18.8	14.2	6.4	-0.1	-6.4	(29)
(30)			MCDWB	-6.0	-1.3	8.3	16.2	23.0	25.2	26.7	24.3	19.7	13.3	4.0	-4.3	(30)
(31)		5%	WB	-10.5	-6.2	0.0	5.5	11.0	16.7	18.8	17.8	13.0	4.8	-1.6	-8.0	(31)
(32)			MCDWB	-8.9	-3.4	5.3	13.6	20.1	24.0	25.4	23.0	18.3	10.3	1.7	-6.2	(32)
(33)		10%	WB	-12.3	-8.0	-1.8	3.9	9.6	15.8	18.1	16.9	11.5	3.3	-3.3	-9.6	(33)
(34)			MCDWB	-11.0	-5.8	2.6	10.8	18.3	23.3	24.1	22.1	16.6	7.8	-0.7	-8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	11.4	12.3	12.4	13.8	13.3	11.9	11.4	11.9	10.5	9.0	8.5	(35)
(36)			MCDBR	10.3	13.3	15.1	18.1	20.2	18.3	16.7	16.1	17.5	15.3	10.7	8.5	(36)
(37)			MCWBR	8.8	10.6	9.8	10.1	9.4	7.2	6.2	6.9	8.5	8.8	7.7	7.1	(37)
(38)		5% WB	MCDWB	10.2	13.0	14.6	16.9	18.3	13.8	13.1	13.2	14.8	14.4	10.4	8.5	(38)
(39)			MCWBR	8.8	10.4	9.6	9.6	8.9	5.9	5.8	6.3	7.6	8.5	7.7	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.280	0.317	0.387	0.481	0.464	0.427	0.388	0.342	0.302	0.276	0.259	(40)	
(41)		taud	2.221	2.196	2.191	2.146	1.914	2.078	2.250	2.345	2.417	2.442	2.312	2.184	(41)	
(42)		Ebn at Noon	766	843	875	846	776	797	823	840	846	819	734	684	(42)	
(43)		Edh at Noon	68	93	115	136	181	155	129	111	92	73	62	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.51	2.64	4.23	5.23	5.83	6.09	5.62	4.84	3.84	2.53	1.50	1.15	(44)	
(45)		RadStd	0.04	0.07	0.13	0.39	0.37	0.42	0.46	0.35	0.29	0.15	0.08	0.06	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page